UNIT - 1

INTRODUCTION

Human-Computer Interaction (HCI) is a rapidly growing field that focuses on improving how people interact with technology. It combines psychology, computer science, and design to create user-friendly interfaces.

HCI has various applications, from business to healthcare. Effective user interfaces can lead to business success, but they can also raise ethical concerns, especially in areas like national security and privacy.

HCI research aims to understand how people use technology and develop better ways for them to interact with it. This includes studying user behavior, creating theories, and conducting experiments.

Challenges in HCI design include creating interfaces that work across different devices and are accessible to everyone, regardless of their abilities. As technology continues to evolve, HCI will play a crucial role in shaping our future interactions

USABILITY GOALS AND MEASURES:

Careful determination of the user community and of the benchmark set of tasks is the basis for establishing usability goals and measures. For each user and each task, precise measurable objectives guide the designer, evaluator, purchaser, or manager.

The ISO 9241 standard focuses on admirable goals (effectiveness, efficiency, and satisfaction), but the following usability measures, which focus on the latter two goals, lead more directly to practical evaluation:

1. **Time to learn.** How long does it take for typical members of the user community to learn how to use the actions relevant to a set of tasks?

2. **Speed of performance.** How long does it take to carry out the benchmark tasks?

3. **Rate of errors by users.** How many and what kinds of errors do people make in carrying out the benchmark tasks? Although time to make and correct errors might be incorporated into the speed of performance, error handling is such a critical component of interface usage that it deserves extensive study.

4. **Retention over time**. How well do users maintain their knowledge after an hour, a day, or a week? Retention may be linked closely to time to learn, and frequency of use plays an important role.

5. Subjective satisfaction. How much did users like using various aspects of the interface? The answer can be ascertained by interview or by written surveys that include satisfaction scales and space for free-form comments.

USABILITY MOTIVATIONS

There's a strong interest in creating user-friendly interfaces due to the poor design of many current systems and the benefits that good design offers. This interest comes from developers of various types of systems, including life-critical, industrial, office, and collaborative systems.

Life-Critical Systems Systems like air traffic control and medical instruments require high reliability and effectiveness, even under stressful conditions. While training time is important, subjective satisfaction is less so for professionals.

Industrial and Commercial Uses Systems used in industries like banking and insurance must be cost-effective and easy to learn. Speed and accuracy are also crucial, but operator fatigue and stress are concerns.

Office, Home, and Entertainment Applications For personal use, interfaces must be easy to learn, have low error rates, and provide satisfaction. Competition is fierce, so users may abandon poorly designed systems.

Exploratory, Creative, and Collaborative Interfaces These systems support intellectual and creative activities, often involving multiple users working together.

Sociotechnical Systems Systems like voting and health support must prioritize trust, privacy, and accountability. They need to be designed for various user levels and provide clear feedback.

Overall, designers of interfaces must consider factors like user expertise, cost-effectiveness, and ease of use to create successful systems.

UNIVERSAL USABILITY

Understanding Users

- Designers must consider the wide range of human abilities, backgrounds, motivations, personalities, cultures, and work styles.
- Different usage profiles (novice, intermittent, expert) and task profiles require varying designs.
- Universal usability is becoming increasingly important due to the growing diversity of users and the push for broader consumer appeal.

Physical Abilities and Workplaces

- Design must accommodate diverse physical abilities, including visual, auditory, and motor impairments.
- Workplace design plays a significant role in user comfort, efficiency, and overall well-being.
- Factors like work surface height, chair adjustability, and lighting can impact user experience.

Cognitive and Perceptual Abilities

- Understanding human cognitive processes, such as memory, problem-solving, and decision-making, is essential for effective design.
- Factors like arousal, fatigue, and mental load influence performance and can impact user interactions.
- Designers should consider the limitations and strengths of human perception and cognitive abilities.

Personality Differences

- Personality traits can influence user preferences, motivations, and interactions with technology.
- The Myers-Briggs Type Indicator is one tool for understanding personality types.
- Designers can tailor interfaces to accommodate different personality styles.

Cultural and International Diversity

- Design must consider cultural differences in language, customs, and preferences.
- Internationalization is crucial for global markets.
- Factors like character sets, reading direction, date and time formats, and cultural metaphors must be considered.

Users with Disabilities

- Design should accommodate users with disabilities, including visual, auditory, and motor impairments.
- Accessibility guidelines and resources are available to help designers create inclusive interfaces.
- Examples of accommodations include keyboard alternatives, color coding, and textual alternatives to images.

Older Adult Users

- Design should consider the physical and cognitive changes associated with aging.
- Adaptations like larger fonts, higher contrast, and easier navigation can improve accessibility for older adults.
- Interfaces should be designed to support the needs and preferences of older users.

Children

- Design for children must consider their developmental stages, limitations, and interests.

- Safety and appropriate content are crucial, especially in online environments.
- Interfaces should be engaging, interactive, and support children's learning and development.

Overall, designers must strive to create interfaces that are inclusive, accessible, and meet the needs of all users. By considering these diverse factors, designers can create more effective, efficient, and enjoyable user experiences.

GOALS FOR OUR PROFESSION :

The field of HCI aims to create interfaces that are intuitive, efficient, and enjoyable for users. To achieve this, it focuses on three primary goals:

1. Influencing Academic and Industrial Researchers

- **Promoting Scientific Research:** Encouraging the use of rigorous research methods, such as controlled experimentation, to understand human-computer interaction principles.
- **Addressing Research Gaps:** Identifying and exploring areas where further research is needed, such as reducing user anxiety, facilitating graceful evolution, and improving input devices.

2. Providing Tools, Techniques, and Knowledge for Commercial Developers

- **Developing User-Interface Building Tools:** Creating software tools that streamline interface development and ensure consistency.
- **Establishing Design Guidelines:** Providing clear guidelines and best practices for interface design.
- **Promoting Iterative Development:** Emphasizing the importance of ongoing evaluation and refinement of interfaces based on user feedback.

3. Raising the Computer Consciousness of the General Public

- **Advocating for User-Friendly Design:** Promoting the benefits of well-designed interfaces and encouraging users to demand high-quality experiences.
- **Educating the Public:** Raising awareness about computer usage and dispelling common misconceptions.
- **Addressing User Concerns:** Helping users overcome fears and anxieties associated with technology.

By achieving these goals, HCI can contribute to the development of more effective, efficient, and enjoyable interfaces that benefit both users and businesses.

Managing Design Processes

Introduction:

The text emphasizes the importance of designing user interfaces that are easy to use and meet the needs of a wide range of people. It highlights the shift from a focus on technical

expertise to accommodating a broader user base with varying motivations and skills. The text also underscores the value of involving users in the design process to create interfaces that are effective and satisfying.

Organizational Design:

Organizations recognize the role of usability for productivity. With the increase of novice users, products with similar functionality can succeed or fail due in part to good usability engineering. A shift in focus towards usability constitutes an organizational change to be managed.

The benefits of this change are; shorter learning times, performance, error reduction. The Return on Investment (ROI) in major corporations is always questioned, the benefits must be made clear. As projects become more complex, the critical specialization in the field of usability and HCI increases. Their are interface-development activities where the ROI for usability analysis is not immediately apparent, but usability of the delivered systems crucial for success.

Some industries, like aerospace, have Human Systems Integration (HSI) requirements that deal with human factors, usability, display design, navigation, and son, while meeting customer requirements. As user-interface design matures, projects grow in complexity, size, and importance where the UI designs take on new perspectives. Usability engineers and user-interface architects (UX) gain experience in managing organizational change. Design is inherently creative and unpredictable.

Method in Recognizing the creative and unpredictable:

 Design is a process, not an end-state and cannot be adequately represented statically

 The design process is non-hierarchical; it is neither strictly bottom-up nor top-down.

 The design process is radically transformational; involves the development of partial and interim solutions

 Design intrinsically involves the discovery of new goals.

The Four Pillars of Design:

A set of fundamentals to assist interface designers optimize usability. These are benefits that are derived from Academic Research and if used properly will assist the designer in creating a Successful Interface.

 Ethnographic Observation provides User-interface Requirements

 Theories and Models provide a means of developing Guidelines Documents & Process

 Algorithms and Prototypes assist the development of User-interface Software Tools which can be used for Rapid-Prototyping.

 Controlled Experimentation provides Expert Reviews & Usability Testing

User Interface Requirements

1. **User Specification**: Clearly define the user group and their specific tasks to ensure software meets their needs.
2. **Target Audience**: Identify who will use the software and how they will interact with it.
3. **Training Relevance**: In learning and development (L&D), understand the audience for training and expected outcomes.
4. **Project Differences**: Tailor e-learning projects based on user needs, e.g., junior sales training vs. leadership development for executives.
5. **Outcome Focused**: Effective software and training design relies on knowing users' goals and backgrounds.

Guidelines, Documents and Processes

1. **Documentation Importance**: Create well-written guidelines to standardize training experiences across different instructors.
2. **Scope of Materials**: Documents can range from short, simple guides to extensive, comprehensive manuals depending on training complexity.
3. **Four E's Framework**: Follow the principles of Education, Enforcement, Exemption, and Enhancement for effective training documents.
4. **Trainer Engagement**: Encourage trainers to discuss and understand guidelines, allowing for clarifications of any questions.
5. **Ongoing Review**: Regularly review and update training materials, ideally once a year, to maintain relevance and accuracy.

User-Interface Software Tools

1. **Client Expectations**: Clients may lack a clear vision of the final product, making it essential to set expectations early.
2. **Cost of Changes**: Making changes post-development can be costly and challenging; avoid this by clarifying designs beforehand.
3. **Early Visualization**: Provide clients with realistic models of the final product before development begins to align expectations.
4. **Training Models**: Use instructional design frameworks like ADDIE to outline training programs clearly from the start.
5. **Financial Considerations**: Effective early communication about training programs helps mitigate financial and time investments.

Expert Reviews & Usability Testing

1. **Pilot Testing**: Conduct small and large pilot tests on software components to identify and fix errors before market launch.

2. **L&D Parallel**: Learning and development experts should similarly test training programs to ensure effectiveness.
3. **Peer Review**: Involve other L&D professionals to review training materials for potential issues or improvements.
4. **Test-Runs for Large Programs**: For extensive training initiatives, conduct small test runs to assess whether learning objectives are met.
5. **Feedback Integration**: Use feedback from testing to refine and enhance the training program before full implementation.

Development methodologies:

Communication Challenges

- Project Failures: Many software projects fail due to poor communication between developers and clients.
- Understanding Needs: Successful developers focus on gathering accurate business requirements.
- User-Centered Design: Focusing on users can save time and money in development.

Successful Developer Practices

1. Gather Requirements: Develop skills to understand client needs.
2. Focus on Design: Prioritize user-centered design during development.
3. Use Methodologies: Apply methods like GUIDE, STUDIO, and OVID.

Rapid Contextual Design Techniques

1. User Interviews: Talk to users in their environments.
2. Analyze Findings: Discuss insights to create useful models.
3. Team Discussions: Collaborate on workflow and issues.
4. Organize Insights: Use affinity diagrams to group ideas.
5. Create Personas: Develop fictional characters to represent users.

Visualization and Prototyping Techniques

1. Visualize Scenarios: Walk through user interactions with personas.
2. Storyboarding: Outline user tasks with visuals.
3. Design User Environments: Build environments based on storyboards.
4. Paper Prototypes: Create simple prototypes for testing.
5. Test on Paper: Use prototypes to gather feedback before finalizing.

By improving communication and using these techniques, development teams can better meet user needs and succeed in projects.

Ethnographic Observation:

Observation of Users

- **Cultural Understanding**: Each user group has its own unique culture, making observation crucial.
- **Ethnographic Methods**: These methods are suitable for observing users in their natural environment.

Ethnographic Approaches

- **Immersion**: Ethnographic methods involve deeply engaging with users by listening, observing, and asking questions to understand their experiences.

Non-Traditional Ethnography

- **Design Focus**: User-interface designers observe users to create designs that improve their lives.
- **Time Constraints**: Designers often have limited time, so observations must be concise and effective.

Intentions for Observation

1. **Influence Design**: Gather insights that directly impact design decisions.
2. **Follow a Valid Process**: Ensure that observations are part of a structured and reliable design process.

Guidelines for an Ethnographic Study

Preparation

1. **Understand Context**: Familiarize yourself with workplace policies and family values in relevant environments.
2. **Review Existing Interfaces**: Learn about the current interface and its history.
3. **Set Goals and Questions**: Define initial objectives and prepare specific questions for your study.
4. **Obtain Access**: Secure permission to observe users or conduct interviews.

Field Study

1. **Build Rapport**: Establish a good relationship with users to encourage openness.
2. **Data Collection**: Observe users and gather both objective/subjective and qualitative/quantitative data.
3. **Follow Leads**: Be flexible and pursue any interesting leads that arise during your visits, documenting your observations.

Analysis

1. **Organize Data**: Compile and arrange collected data into a structured format (e.g., databases).
2. **Quantify and Summarize**: Use statistics to quantify findings and summarize key insights.
3. **Interpret Data**: Reduce complexity and draw meaningful conclusions from the data.

4. **Refine Goals**: Revisit and adjust your initial goals based on insights gained.

Report

1. **Consider Your Audience**: Tailor your findings for multiple audiences and their specific interests.
2. **Prepare a Report**: Write a comprehensive report that presents your findings clearly and effectively.

By following these guidelines, you can conduct an effective ethnographic study that provides valuable insights for design and development

Participatory Design:

Participatory design in Human-Computer Interaction (HCI) is a collaborative approach that actively involves users in the design process. This method empowers users to contribute their insights, experiences, and feedback, ensuring that the final product aligns with their needs and expectations. By treating users as co-designers, participatory design fosters a more user-centered approach to creating technology and interfaces.

Key Aspects of Participatory Design:

1. **User Involvement**: Users collaborate in the design process, providing valuable insights.
2. **Co-Designing**: Users engage in brainstorming and design activities to influence outcomes.
3. **Democratic Approach**: Emphasizes equal participation from all stakeholders for diverse perspectives.
4. **Iterative Feedback**: Involves creating prototypes that users test and provide feedback on.
5. **Focus on Context**: Users share insights about their environments and workflows to inform design.

Scenario Development :

Scenario development is a technique used in Human-Computer Interaction (HCI) to create detailed narratives that describe how users interact with a system in specific contexts. These scenarios help designers understand user needs, workflows, and the environment in which the product will be used.

Scenarios

 “Day-in-the-life” helpful to characterize what happens when users perform typical tasks. Analogs and metaphors

 Describes novel systems

 Write scenarios of usage and then if possible, act them out as a form of theater, effective with multiple user cooperation.

Useful applications

 Control rooms, cockpits, financial trading rooms and other places where people must coordinate

Social Impacts

Interactive systems often have a dramatic impact on large numbers of users. To minimize risks, a thoughtful statement of anticipated impacts circulated among stakeholders can be a useful process for eliciting productive suggestions early in development.

Examples of a social impact statement

- Describe the new system and it's benefits

o Convey the high-level goals of the new system

o Identify the stakeholders. o Identify specific benefits

- Address concerns and potential barriers

o Anticipate changes in job functions.

o address security and privacy issues

o Describe accountability and responsibility for system misuse and failure.

o Avoid potential biases.

o weigh individual rights o assess trade-offs between centralization and decentralization o Preserve democratic principles

o ensure divers access o Promote simplicity.

- Outline the development process

o Present an estimated project schedule

o Propose a process for making decisions discuss expectations of how stake holders will be involved.

o Recognize needs for staff, training and hardware.

o propose a plan for backups of data

o Outline a plan for migrating to the new system.

o Describe a plan for measuring the success of the new system

Legal Issues

Privacy

- Concerns arise over sensitive data (medical, legal, financial) that may be accessed or tampered with illegally.

Safety and Reliability

- User interfaces for critical systems (aircraft, automobiles, medical devices) must ensure safety, as failures can lead to life-or-death situations.

Patent Protection for Software

- Developers may patent software to recover costs and generate profit from their projects, raising questions about intellectual property rights.

Piracy and File Sharing

- Legal issues regarding users' rights to store, share, and own digital content, including the legality of file-sharing practices.

Freedom of Speech

- Questions arise about users' rights to express controversial or offensive opinions via email or online platforms.

Localization

- Compliance with various legal standards across different jurisdictions (international, federal, state, and municipal) is essential for software deployment.

These legal issues highlight the complexities that arise as user interfaces become increasingly integrated into various aspects of life and business.

Usability Testing and Laboratories :

The usability-test report provided supportive confirmation of progress and specific recommendations for changes.

 Usability testing not only sped up many projects, but also produced dramatic cost savings.
 Usability tests are designed to find flaws in user interfaces

Usability testing comes in many different flavors and formats. Most of the current research demonstrates the importance of testing often and at varied times during the design cycle. The purpose of the test and the type of data that is needed are important considerations. Testing can be performed using combinations of these methods as well.

 Paper mock-ups and prototyping.

o A test administrator plays the role of the computer by flipping the pages while asking a participant user to carry out typical tasks.

o Inexpensive, rapid, and usually productive.

 Discount usability testing.

o Quick and dirty approach to task analysis, prototype development, and testing o Widely influential because it lowers the barriers to newcomers.

o Advocates point out that most serious problems are found with only a few participants, enabling prompt revision and repeated testing.

 Competitive usability testing.

o Competitive testing compares a new interface to previous versions or to similar products from competitors.

 Universal usability testing.

o This approach test interfaces with highly diverse users, hardware software platforms, and networks.

o This will result in the creation of products that can be used by a wider variety of users.

 Field tests and portable labs.

o This testing method puts new interfaces to work in realistic environments or in a more naturalistic environment in the field for a fixed trial period.

 Remote usability testing.

o Since web-based applications are available internationally, it is tempting to conduct usability tests online, avoiding the complexity and cost of bringing participants to a lab.

o This makes it possible to have larger numbers of participants with more diverse backgrounds, and it may add to the realism, since participants do their tests in their own environments and use their own equipment.

Usability Labs

 A typical modest usability laboratory would have two 10-by-10-foot areas, divided by a halfsilvered mirror.

Step-by-Step Usability Guide

 Plan Analyze Design Test and Refine